Krittika Summer Projects 2025

Bangs, blips, or background

Decoding GRB signals

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Abstract

Gamma-Ray Bursts (GRBs) are among the most energetic and enigmatic astrophysical phenomena observed, requiring precise and robust techniques for their detection and analysis. In this project, we develop and implement a comprehensive pipeline for identifying GRBs using data from the Cadmium Zinc Telluride Imager (CZTI) onboard AstroSat. The analysis begins with preprocessing of event-mode FITS data, followed by signal binning, smoothing, masking of instrumental effects such as the South Atlantic Anomaly (SAA), and detrending to enhance transient visibility. A peak detection algorithm based on the N-sigma method is applied, followed by a stringent multi-quadrant coincidence check to distinguish genuine astrophysical signals from instrumental noise and charged particle events. Spectral and temporal characteristics are validated using spectrograms and Veto detector analysis. The final algorithm successfully identified and validated a candidate GRB exhibiting a hard energy spectrum, consistent multi-quadrant detection, and significant Veto response—hallmarks of a bright, high-energy event. This study demonstrates the effectiveness of statistical techniques combined with instrument-specific knowledge in automating GRB detection and paves the way for real-time transient identification in future high-energy missions.

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1 Introduction

1.1 Gamma-Ray Bursts (GRBs)

1.1.1 Definition and Overview

Gamma-ray bursts (GRBs) are short-lived bursts of gamma-ray light and are one of the brightest, most powerful explosions that we've ever observed in the universe. These are brief explosions but emit gamma rays at an highly intense level; which is the highest-energy form of electromagnetic radiation. These emissions can last from milliseconds to minutes, depending on their source and because of the high invariance of their behaviour, studying their light curves becomes a challenging activity. However, studying these persistent bursts gives us valuable information about the source they are emitting from and provides us with insights into the physical conditions, composition, and dynamics of some of the most extreme environments in the universe.

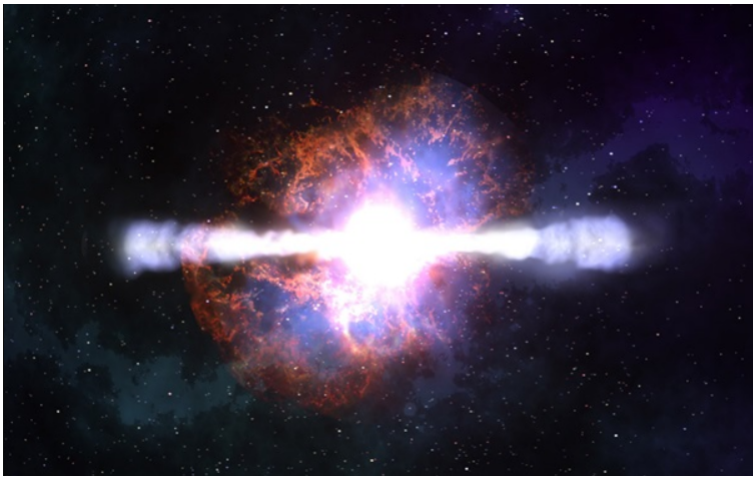


Figure 1: NASA's Goddard Space Flight Center shows the emergence of GRB from the collapse of a black hole. [1]

Although GRBs are one of the brightest phenomena occurring in the universe, it is not always easy for us to spot them. This is because unlike studying optical or radio waves, observing gamma rays serves as a challenge for us. We are, however, able to detect them with greater ease using specially designed space-based instruments. These operate beyond the Earth's atmosphere, where gamma rays can reach detectors without being absorbed by the atmosphere. There are certain longer lasting emissions which linger around after a GRB takes place, this is known as afterglow. These afterglows occur when material in the jets interact with surrounding gas [2].
While GRBs usually only last seconds or at most minutes, these afterglows can stay around for a long time period which makes it a valuable source to study. Early searches for this afterglow were unsuccessful, largely because it is difficult to observe a burst's position at longer wavelengths immediately after the initial burst. When we usually detect a burst, it takes millions of years for the light to travel that long of a distance and this is when afterglows help us learn more about the object in question [3]

1.2 Classification of GRBs

The classification of GRBs is mainly done in three parts. We have short GRBs, where the initial burst of gamma rays lasts less than two seconds, long GRBs, which last longer than 2 seconds and Ultra long GRBs which can last up to tens of thousands of seconds [2].

1.2.1 Short GRBs

These GRBs are thought to have originated from mergers of a binary neutron star or collision between a neutron star and a black hole which led to the formation of a black hole while also releasing vast amounts of energy. They often lack an associated supernova. Such mergers are likely to produce a kilonovae, which is an expanding luminous afterglow of radiation emitted after one of the collisions mentioned above [4]. This was conclusively established in 2017, when short GRB 170817A co-occurred with the detection of gravitational wave GW170817, a signal from the merger of two neutron stars [5].

1.2.2 Long GRBs

Most of the events that are observed do have a duration of more than 2 seconds which makes them long GRBs. These GRBs are linked to the explosive deaths of massive stars; in order words, supernovae. These are also formed from rapid star formation. These GRBs usually last tens to hundreds of seconds. Long GRB afterglow observations, at high redshift, are also consistent with the GRB having originated in star-forming regions [3].

1.2.3 Ultra Long GRBs

Some bursts manage to last over hours which may been the result of either newborn magnetars or the collapse of a supergiant star. We have till now only been able to identify a few of them which might be because of the poor sensitivity of our instruments towards long duration events. The longest ever GRB recorded is GRB 111209A which lasted at least around 7 hours and has been sourced to be originated from a blue supergiant star.

1.3 Source of origin

GRBs originate from different catastrophic astrophysical events depending on their type. Long GRBs are usually emitted due to the collapse of massive stars. The most widely accepted explanation for how long-duration GRBs form is called the collapsar model. In this model, the core of a very massive, fast-spinning star collapses into a black hole at the end of its life. As the core collapses, the surrounding material falls inward and forms a dense, spinning disk around the black hole. This process creates two powerful jets that shoot out from the poles of the star. These jets punch their way through the outer

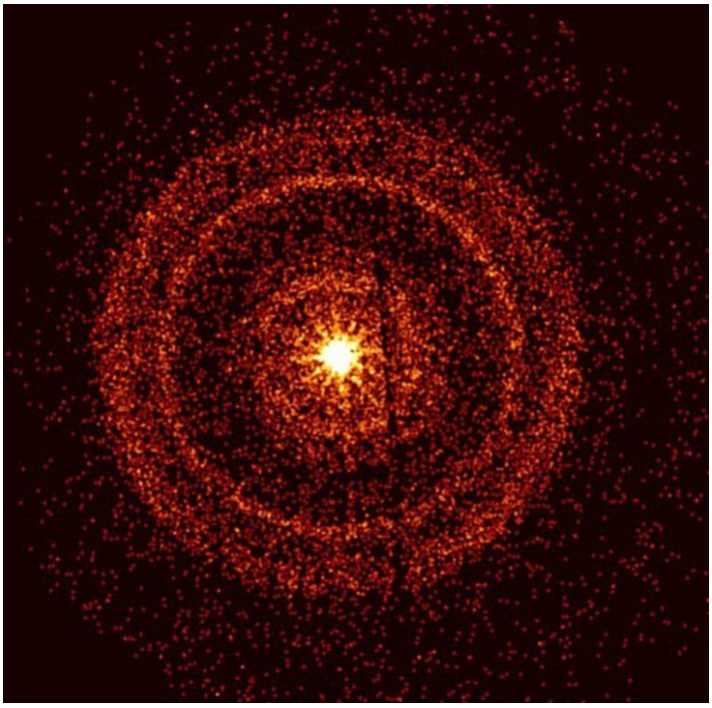


Figure 2: Swift captured the afterglow of GRB 221009A about an hour after it was first detected reaching Earth on October 9, 2022.[6]

layers of the star and, once they break free, they release a burst of gamma rays — what we detect as a gamma-ray burst [3].
For short gamma rays, usually the merging of two stars in a binary system is considered as a source of origin. According to this model, the two stars slowly rotate around each other until they collapse into a black hole. The infall of matter into the new black hole produces an accretion disk and releases a burst of energy, analogous to the collapsar model. Apart from these models, other models are also hypothesized and followed like using a magnetar instead of a star and using a star and a black hole merger.

1.4 Astrosat

1.4.1 Introduction to astrosat

AstroSat is the first dedicated Indian astronomy space telescope aimed at studying celestial sources in X-ray, optical and UV spectral bands simultaneously. It was launched on 28 September 2015 and follows a near-equitorial path. One of the unique features of the AstroSat mission is that it enables the simultaneous multi-wavelength observations of various astronomical objects with a single satellite. It has aboard a total of six instruments in its payload which helps it observe the universe in different wavebands at the same time. The main objectives of AstroSat are monitoring the night sky for search of new transients, conducting broadband spectroscopic studies of X-ray binaries, AGN and SNRs and studying periodic and non-periodic variability of X-ray sources [7].

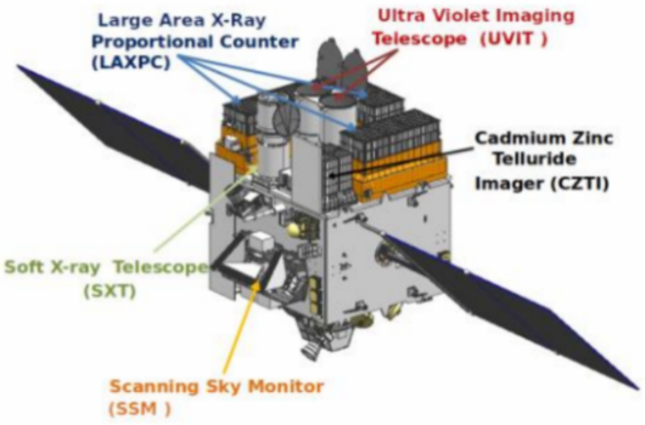


Figure 3: AstroSat instrument [8]

1.4.2 South Atlantic Anamoly

The instrument AstroSat faces a major obstacle upon every revolution around the Earth. It passes around a region near the South Atlantic Ocean at about a height of 700 km around every 98 mins. This region is heavily affected by the inner belt of the Van Allen Radiation belts which dips a little closer to the surface of the Earth creating a region of high radiation consisting of protons and electrons with high energies. This zone with enhanced radiation is called the South Atlantic Anomaly or SAA. AstroSat makes about 14.5 revolutions of the Earth every day, causing it to turn off its instruments every time it passes through this region. Not turning off the instruments or keeping them at the same power can lead to issues such as high data corruption and risk of damaging sensitive electronics. We go on to apply “masking” later in this report to deal with the empty timestamps we have in our files caused due to SAA.

1.5 CZTI

The Cadmium Zinc Telluride Imager (CZTI) is a high energy, wide-field imaging instrument on AstroSat. CZTI was the first astronomical payload to be switched on AstroSat. It provides observations in the hard X-ray energy range. The detectors of CZTI cover an energy range from 10 keV to >200 keV. CZTI is sensitive in the hard X-ray band (20–150 keV), it analyzes light curves and spectra of known X-ray binaries to study compact object accretion, jet formation, and pulsar emission. In addition to the capability of extending the hard part of the energy range for studying X-ray binaries and AGNs it also has the capability to detect Gamma ray bursts. The CZTI is fabricated in 4 identical, independent quadrants which are joined together in the final configuration. These quadrants are used to improve the design’s safety and redundancy. In each quadrant, 16 CZT modules are used and each module is pixelated. So each quadrant contains a 64x64 element coded mask and a detector array of the same number of pixels, with the mask patterns of adjacent quadrants being rotated by 90 degrees relative to each other. So each detector module has 256 pixels. CZT

modules perform best in the temperature range 0 to 15 degrees celsius and hence the heat generated by the detector assembly is drained out continuously by a radiator panel assembly [9] [10].

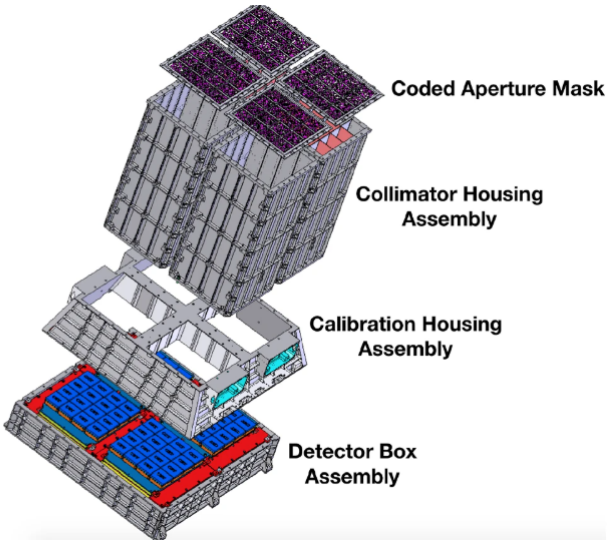


Figure 4: CZTI detector onboard the AstroSat intrument [11]

1.6 Charged particles

One of the key challenges in detecting gamma-ray bursts (GRBs) from space is the interference caused by charged particles, such as cosmic rays and solar energetic particles. These high-energy particles constantly interact with the detectors aboard satellites and can produce signals that mimic genuine astrophysical events [**nasa:grbnasa:grb**]. These particle-induced signals are often indistinguishable in shape and timing from actual GRB signals, especially the short-duration ones. For example, when cosmic rays strike the detector, they may produce brief, high-intensity events that closely resemble short GRBs. In the Cadmium Zinc Telluride Imager (CZTI) on AstroSat, charged particle events are filtered using a combination of energy thresholding, spatial coincidence checks, and comparisons across multiple detector modules. Despite these strategies, identifying and eliminating particle-induced events remains an active area of research, especially when studying faint or borderline detections. In this project, we try to create and implement our own algorithm to distinguish GRBs from these charged particle events [3].

2 Computational Work

2.1 FITS files

2.1.1 Data headers

We work with FITS files to access and analyze datasets obtained from the Cadmium Zinc Telluride Imager (CZTI). FITS stands for Flexible Image Transport System and is an open standard defining a digital file format. It is used for storage, transmission and processing of data which is usually stored in the form of 2D arrays and tables. It is widely adopted in the field of astronomy because it is flexible, self-describing, and capable of storing complex scientific data along with metadata in a structured way. We start by opening our FITS file for the dataset

`AS1C09_006T01_9000006294_47011cztM0_level2_quad_clean`

using the AstroPy package. We use the function with fits.open from astropy.io.fits that opens a FITS file and returns an HDUList. An HDU list is a Header Data Unit List which contains a list-like object containing one or more HDUs. Now HDUs stand for Header Data Unit and is a fundamental block in the formation of a FITS file. It is the highest level component of the FITS file structure, consisting of a header and (typically) a data array or table. After the above open call, hdul[0] is the primary HDU, hdul is the first extension HDU or the BinTable HDU. We use the primary HDU to access the main detector information. We analyze this header to understand how we have to work with the data. Firstly, we print out the columns from the primary header.

```
[array([[name = 'TIME'; format = 'D'; unit = 's',
        name = 'CZTSECCNT'; format = 'D'; unit = 's',
        name = 'CZTNTICK'; format = 'I'; unit = 'micro-sec'; bscale = 1; bzero = 32768,
        name = 'PHA'; format = 'I'; unit = 'counts'; bscale = 1; bzero = 32768,
        name = 'DetID'; format = 'B', name = 'pixID'; format = 'B',
        name = 'DETX'; format = 'B', name = 'DETY'; format = 'B',
        name = 'veto'; format = 'I'; unit = 'counts'; bscale = 1; bzero = 32768,
        name = 'alpha'; format = 'B'; unit = 'counts',
        name = 'PI'; format = 'I'; bscale = 1; bzero = 32768,
        name = 'ENERGY'; format = 'E']], dtype=object)]
```

Figure 5: Primary data columns

We observe different columns like ‘TIME’, ‘DetID’ and ‘ENERGY’. On printing the time column, we notice that the time is not taken uniformly. This is because the given file is an event file. The TIME column tells us about the arrival time of each photon event. It does not provide us with the counts but only the information like when does an event occur and with what energy. In order to analyse this further, we wish to convert this to a counts file. We do this by binning our data into required time bins in the subsequent sections. We also note that the DetID column here gives us the information about what module the event was observed on. We divide it by 4 to get the true value of what quadrant this event was recognized in. As we bin our time column, we also categorize the quadrant the counts belonged to and what energy they possessed.

2.2 Data reduction and Preprocessing

2.2.1 Signal binning

We first begin by importing all the required libraries in our environment. The most important and used libraries or packages are astropy, numpy and matplotlib.pyplot. Let's take a look at each one of them separately.

- **Astropy**

  It is primarily used in astronomy and astrophysics. It offers a broad set of tools for working with units, coordinates, times, data tables, and common astronomical file formats like FITS. The core subpackages that we will be using for this project are astropy.io.fits and astropy.table. "astropy.io.fits" helps with importing and analyzing the data present in our files because we carry on any operation. Whereas "astropy.table" is used later to view the items of the VetoSpec array [12]

- **Numpy**

  We use numpy for our mathematical and numerical operations while managing arrays. While working with data files similar to ours, one might find the size of the dataset to be vast. To avoid rigorous and time consuming steps while handling these datasets, we use numpy. It efficiently handles all these files and provides fast array operations. We also find some very useful inbuilt functions in this library like np.max and np.where. It is most commonly used with other libraries like astropy and matplotlib [13].

- **Matplotlib.pyplot**

  This library is mainly used for plotting and visualising data we have on curves. Using this we can plot line graphs, scatter plots, histograms, heatmaps, etc. as per our wish.

We now go on to analyze the data from the primary HDU or the main detector to search for the appearance of GRBs and/or charged particles. We set the compounds such as 'Time', 'Quadrants' and 'Energy' from the primary header. We notice that we don't have a column of counts. This is because all these rows show us the list of events, not the number of events. To calculate the counts per time bin and make the time uniform, we bin our time row using some time_bins. We use the function np.histogram from the numpy library to do this. Let us take a look at what this function does.
np.histogram - np.histogram is a function from the NumPy library in Python that computes the histogram of a dataset. A histogram is a way to count how many values fall into specific ranges called bins. It's often used to visualize or analyze the distribution of numerical data.
For now, we continue with choosing a time bin of 1 seconds, we will later alter this to analyze different graphs. We can obtain the overall counts or use quadrants to obtain quad-wise counts. We go ahead to plot the curve using plt.plot. We obtain the given curve:

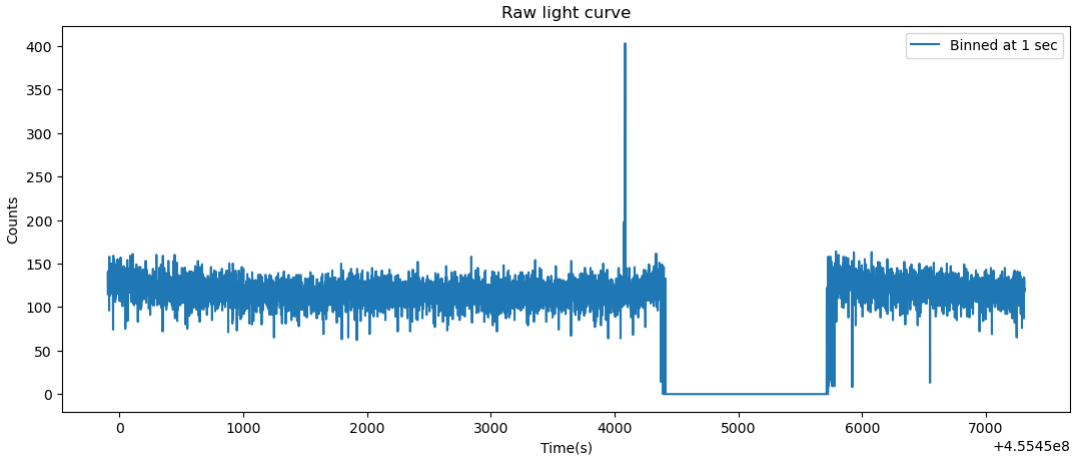


Figure 6: Lightcurve obtained by plotting raw data with time

This gives us a fair estimate of what we're gonna be working with. We take into consideration a couple of points from this lightcurve; from the noisy data to the edge effects and also the SAA region dip.

2.2.2 Masking

Before moving onto the analysis we notice that the SAA region shows zero data, which isn't great for further analysis. This is because when you detrend a light curve or time series and replace SAA intervals with 0, you're creating artificial dips in the data that can distort further analysis. These "zeros" are not real data and are just placeholders for "no data" or "corrupted data" and treating them like actual measurements would skew statistical calculations, like mean, variance, or even FFTs. We apply a mask to get rid of this false data. We pick a window of size 50, check for its standard deviation to be less than 1 and mask the regions where this condition holds true. We obtain the given graph:

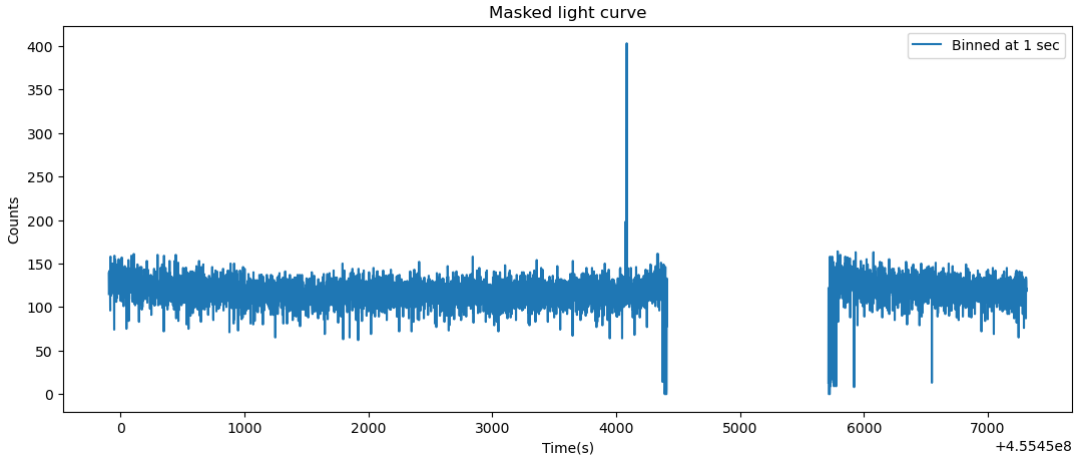


Figure 7: Lightcurve obtained by masking the SAA region

2.2.3 Smoothing

We notice that the curve is too noisy and needs to be smoothed a bit before we move further. We look at three filters for smoothing of a curve. Let us take a closer look at each of them one by one.

- **Median filter** The SciPy library in python provides us a median filter function which directly bins the data into required kernels and calculates their median. This method is great for removing noise, especially spiky outliers [14]
- **Gaussian filter** This filter applies a Gaussian function as a weighting kernel over a one-dimensional data array. Its primary purpose is to reduce noise and suppress high-frequency variations in the signal without significantly distorting the underlying trend. The Gaussian function is defined as:

  $$G(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  Here, x is the distance from the center of the kernel and sigma is the standard deviation controlling the width of the filter. This filter convolves this Gaussian kernel with the data, assigning higher weights to values near the center of the window and progressively lower weights to those further away. [15]
- **Savitsky-Golay filter** Savgol filter stands out due to its ability to smooth data while preserving its important features. It fits successive sub-sets of adjacent data points with a low-degree polynomial (where the degree of the polynomial is also a hyperparameter) using the least square regression. The value of the polynomial at the central point of the window is then taken as the smoothed value. This process is repeated for each point in the data set, resulting in a smoothed signal.

  We can use different combinations of window sizes and polynomial degrees which fits our needs the best. We choose a polynomial order of 2 to prevent overfitting of the curve and preserve the shape of the features like peaks and valleys. A small window size smooths the data but may not capture the overall trend well, especially for signals with higher frequency components [16].

We choose to go with the median filter. This decision was based on its ability to remove random noise while preserving sharp, transient features—a key requirement for accurate GRB detection. We get the following curve applying all the filters.
We notice how the light curve is bent with different slopes at different points. We wish to even these out in order to observe our curve more clearly. This is where detrending comes in.

2.2.4 Handling edge effects

We now have a clean smoothed curve, however we notice the huge outliers near the SAA region. We will apply another mask to ignore or avoid these edge counts near the SAA region for our final analysis. We find the region around which we transition from

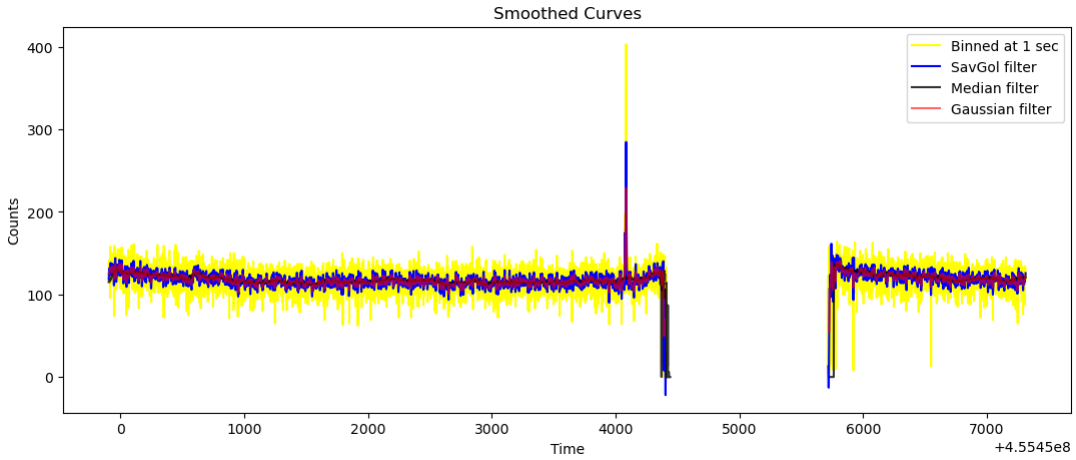


Figure 8: Masked lightcurve obtained by median filtering

NaN to valid or valid to NaN and then mask the buffer regions around each transition. This essentially turns the value of the edge artifacts to be NaN and gives us a cleaner view of the curve.

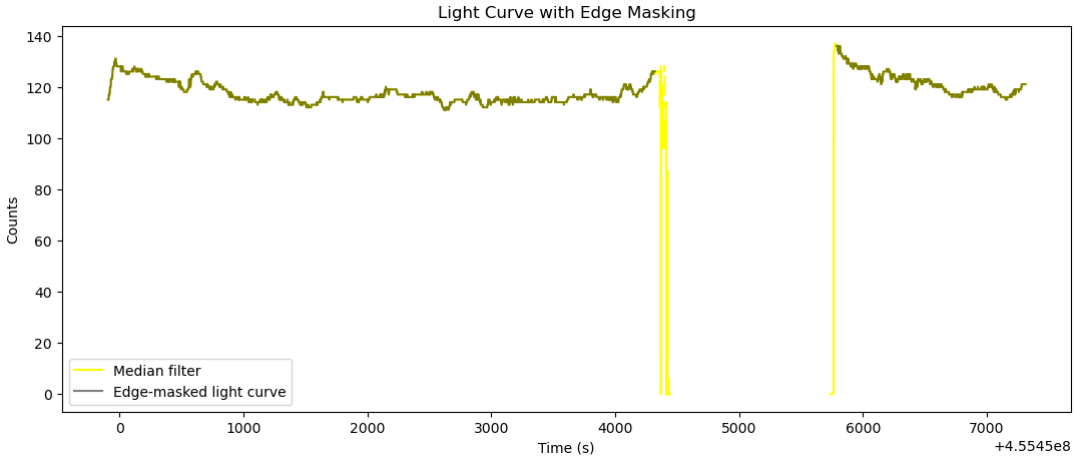


Figure 9: Lightcurve with edge effects removed compared with smoothed lightcurve

2.2.5 Detrending the data

Detrending is the process of subtracting the raw data of the curve from its smoothed counterpart. It removes long-term trends or low-frequency variations from data, so we can better analyze the underlying patterns or short-term fluctuations like short GRBs. If the data is not detrended (as shown above), then slow upward or downward drifts can hide sudden changes we notice in our graph, which might either be due to GRBs or due to some random transient. We follow the standard procedure and subtract the smoothed curve from the raw data curve. We get the following results:
Before moving further to analyze peaks, we look at the data from various quadrants and energy bands and make a note of the inferences.

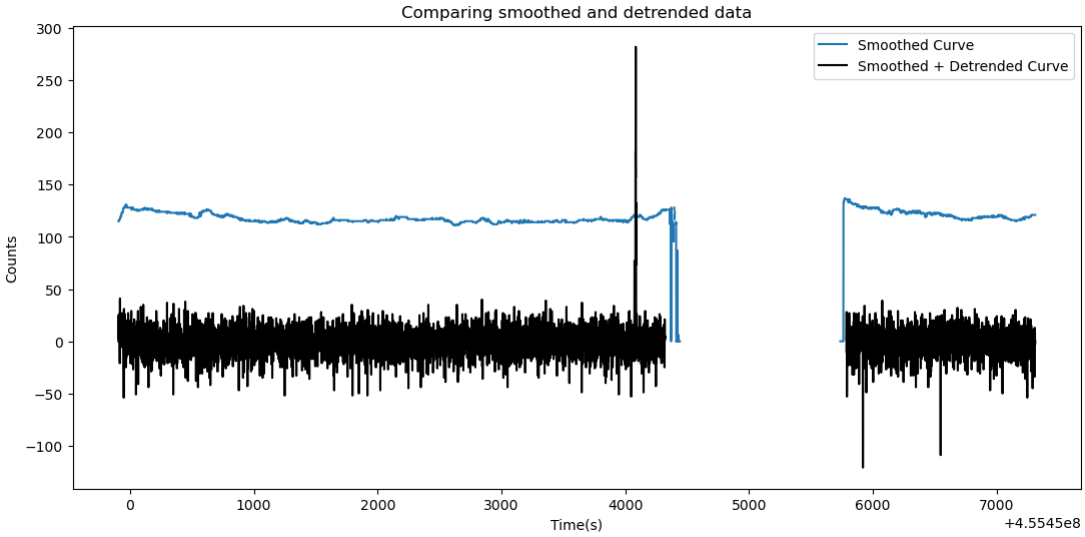


Figure 10: Smoothed lightcurve compared with the detrended lightcurve

2.3 Analyzing data

2.3.1 Quadrant Analysis

Now that we have our detrended smoothed curve, we go on to look at how it behaves with different quadrants and different energy bands. First we look at the different quadrants from this curve and their counts. To do so, we classify the time according to the ones recorded by a particular quadrant and then use the np.histogram function to find the counts in those time periods.

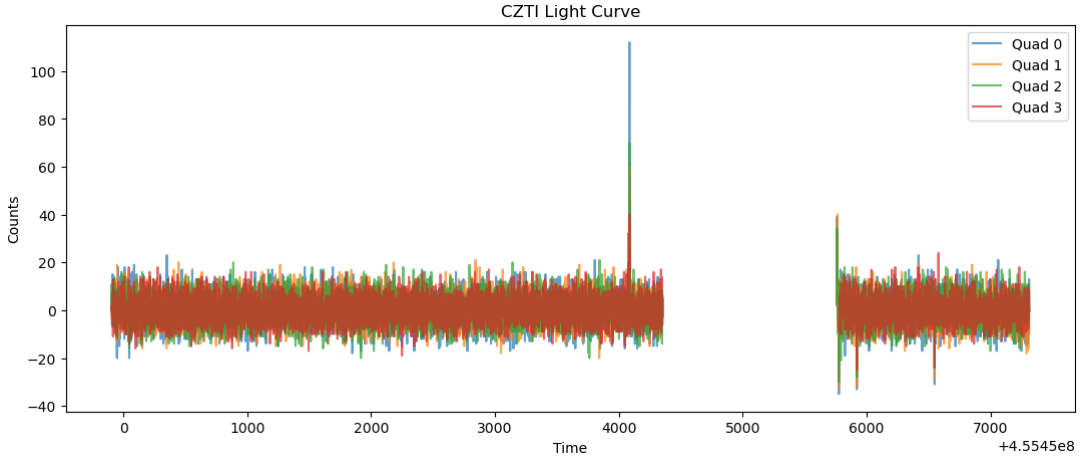


Figure 11: All quadrants plotted in a single plot

This tells us that the data is consistent across all the quadrants and the background level is also symmetric. We notice a strong spike around the time 455484000 + 4000 s. We plot all the quadrants separately to confirm its presence.
We plotted the CZTI light curves for all four quadrants combined and separately to investigate the temporal structure of the detected signal. A significant transient is observed at

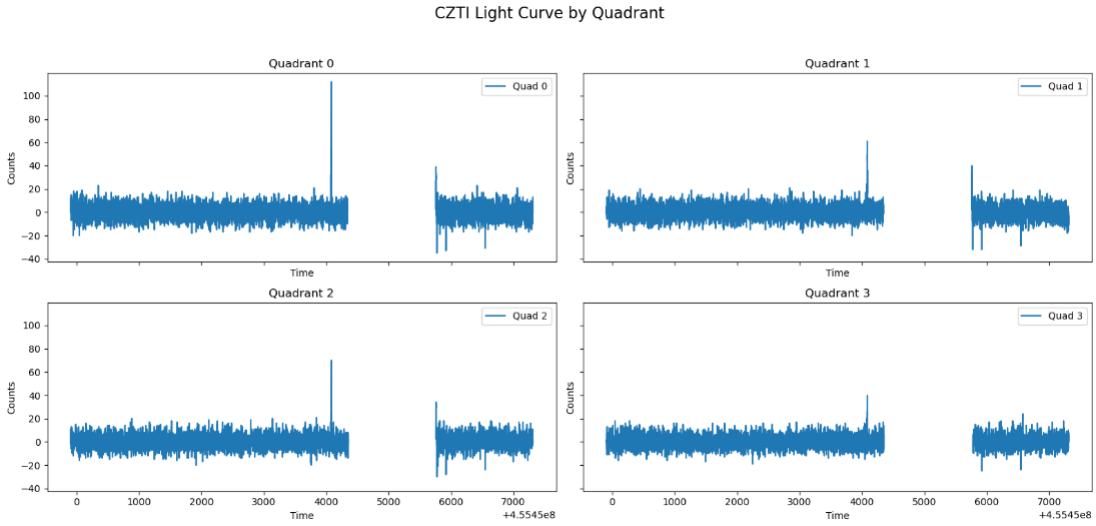


Figure 12: All quadrants plotted separately for the lightcurve

T+4000 s, visible consistently across all quadrants, suggesting an astrophysical origin rather than instrumental noise. Quadrant 0 exhibits the highest signal amplitude, potentially indicating its alignment with the source direction. A secondary feature appears after the SAA gap which might've arised due to the edge artifacts.
Now we apply the same principle to look at the graphs of different energy bands belonging to different quadrants. We do this for multiple quadrants and multiple energy bands and analyze the results.

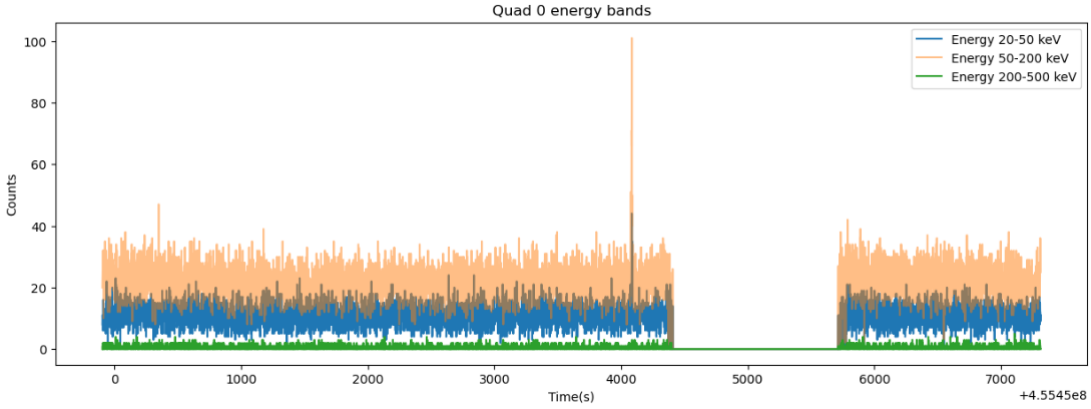


Figure 13: Light curve for quadrant 0 for all energy bands

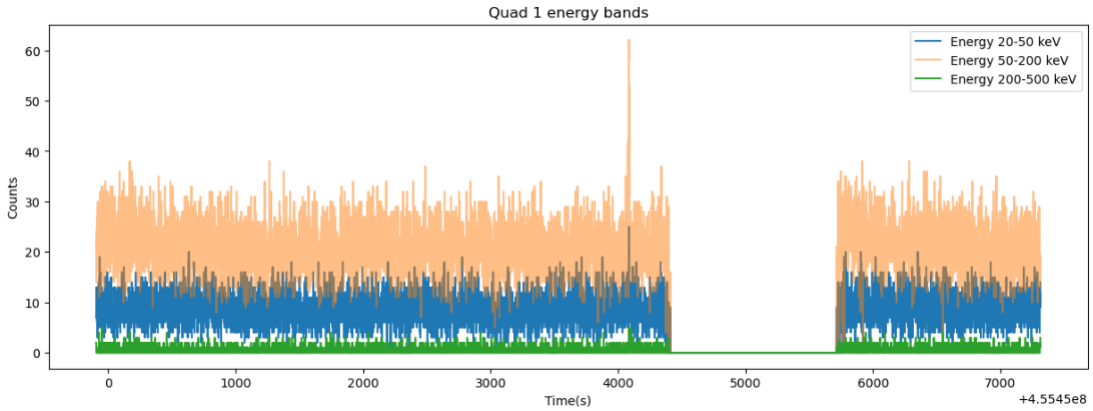


Figure 14: Light curve for quadrant 1 for all energy bands

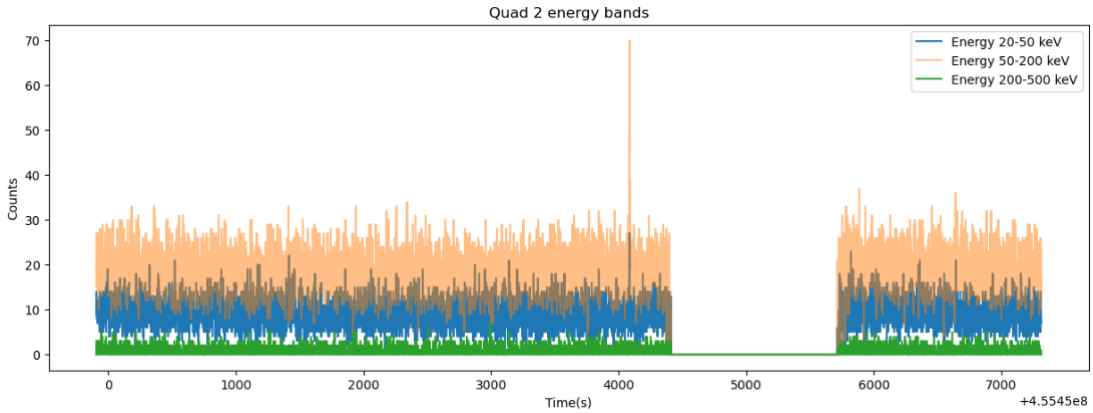


Figure 15: Light curve for quadrant 2 for all energy bands

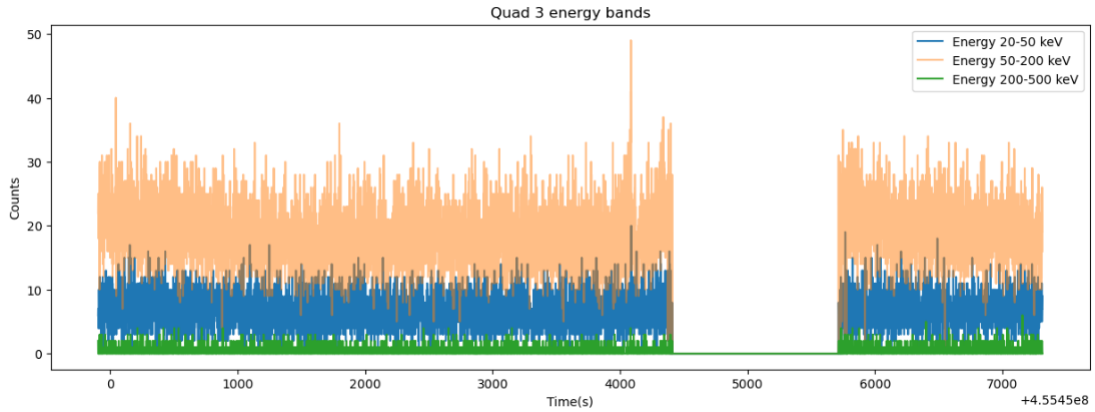


Figure 16: Light curve for quadrant 3 for all energy bands

We notice that the counts in the energy band 50-200 keV is consistently high in all the quadrants. This region is sensitive to typical GRB (Gamma-Ray Burst) prompt emission, pulsars, or magnetars. This is because they are dominated by non-thermal processes, such as synchrotron radiation or Compton scattering, which are often seen

in phenomena like pulsars and Gamma-Ray Bursts. GRBs are typically not visible in the energy band 20-50 keV as they often show thermal emission, weaker bursts and are sensitive to softer sources like X-ray binaries, background, or diffuse gas.

2.3.2 Energy analysis

We now look at the different energy bands in more depth to understand the characteristics of the dataset. First we go on to plot the different energy bands for all quadrants.

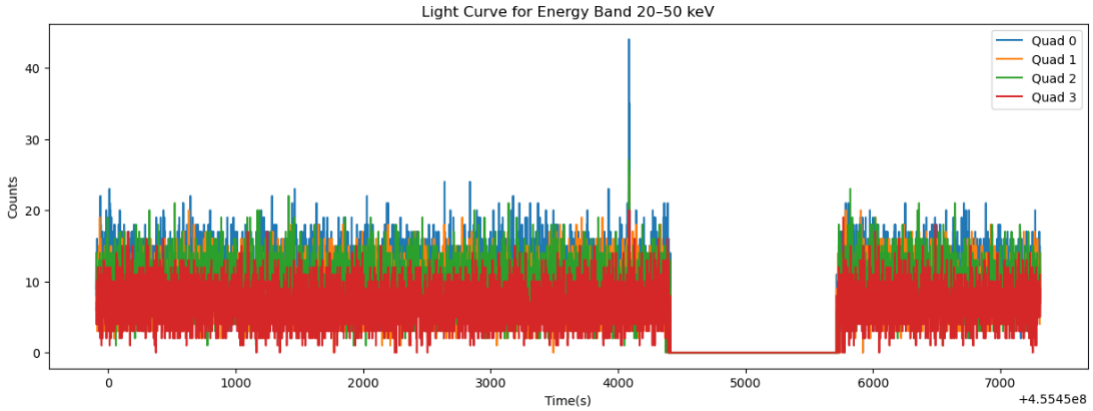


Figure 17: Light curve in energy range 20-50keV for all quadrants

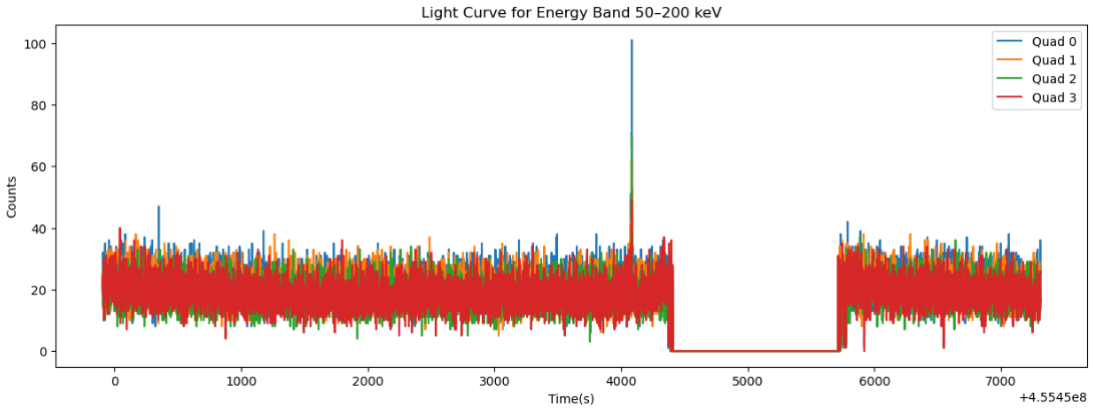


Figure 18: Light curve in energy range 50-200keV for all quadrants

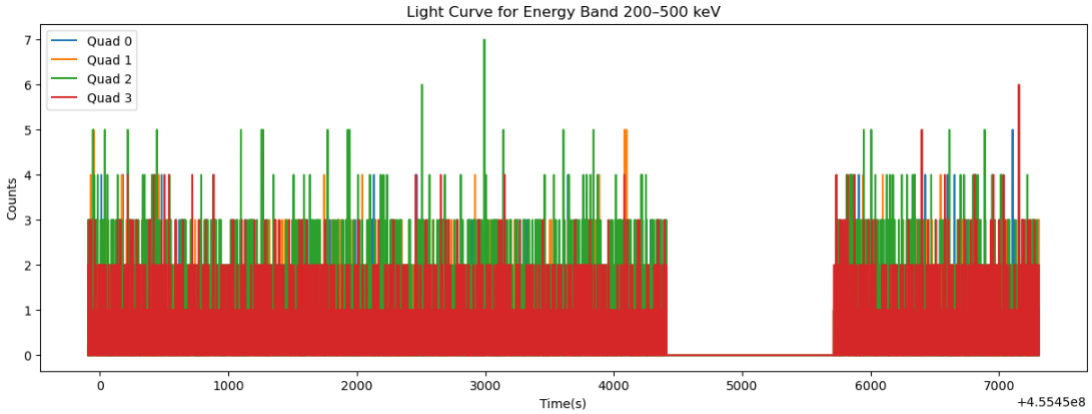


Figure 19: Light curve in energy range 200-500keV for all quadrants

We also plot the counts vs time graph for all the quadrants in different energy bands. We get:

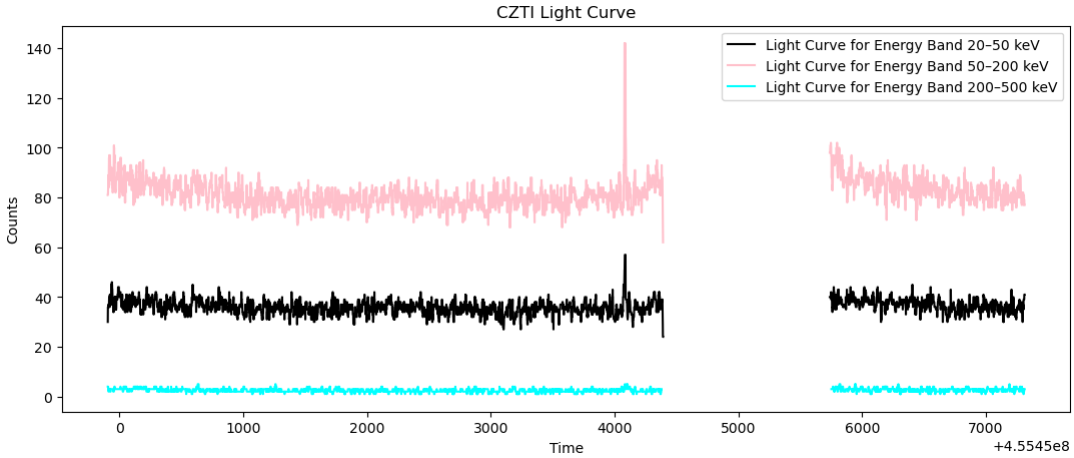


Figure 20: Light curve for all quadrants for different energy bands

We notice that the highest counts for the energy band 50-200 keV is from quadrant 0 and 2. We also note the high energy spike noticed in all the quadrants for different energy bands. Since it is most prominent in the 50-200 keV band it is most like a transient. Moreover, all 4 quadrants show similar patterns, especially during the transient which suggests it's not noise or instrumental, but a real astrophysical signal.

Veto detector analysis

We now examine the Veto detector data to investigate its energy characteristics and assess its role in background rejection. We go on to plot the energy curves in the data detected by veto detectors.

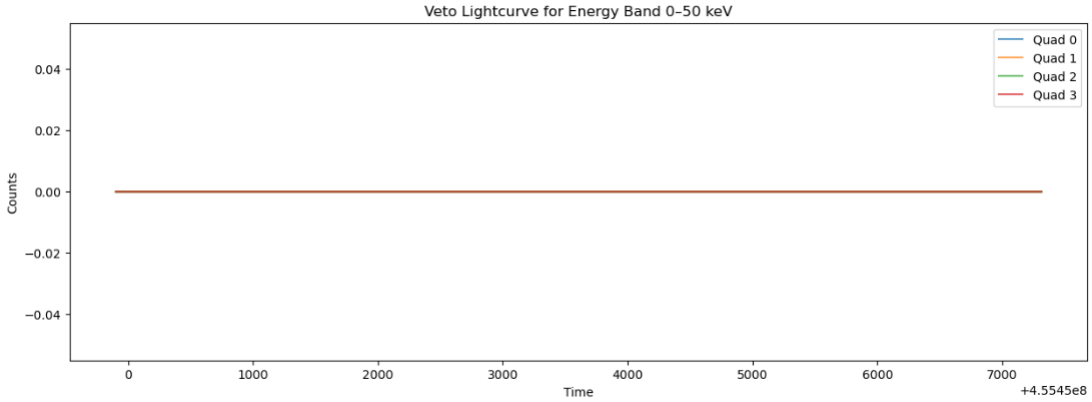


Figure 21: Quadrant wise energy profile for band 0-50keV

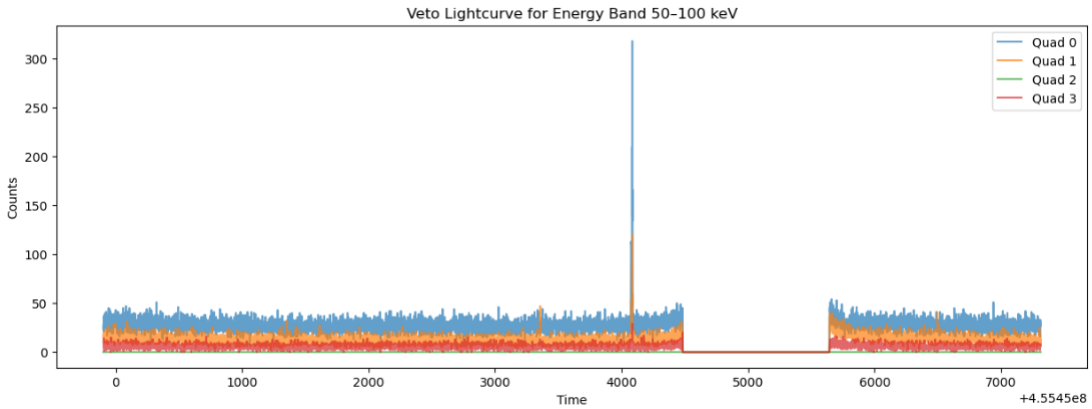


Figure 22: Veto curve in energy range 50-100keV for all quadrants

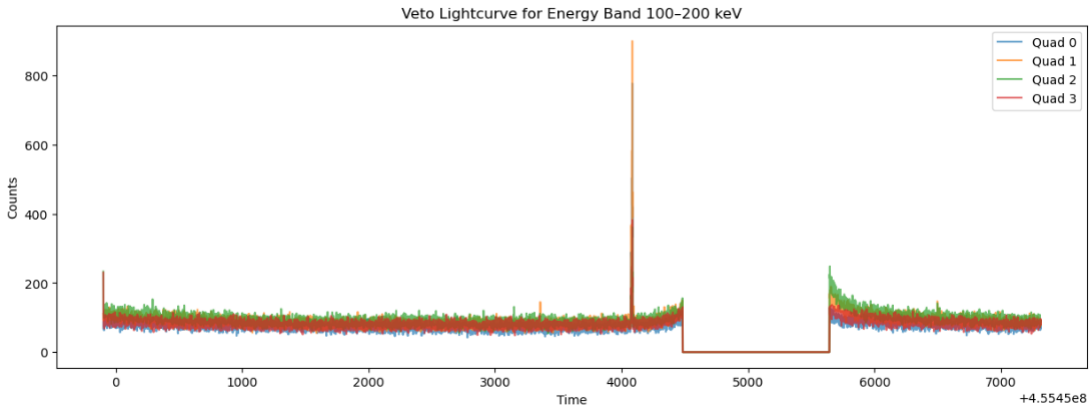


Figure 23: Veto curve in energy range 100-200keV for all quadrants

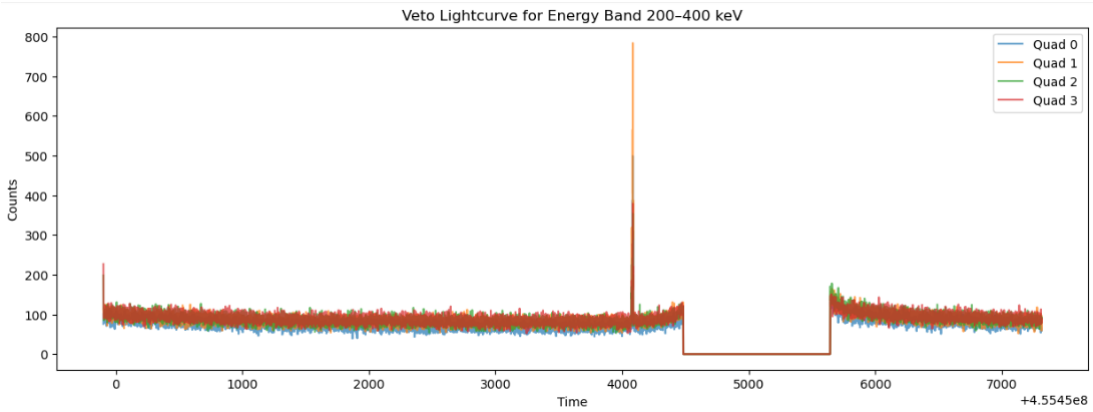


Figure 24: Veto curve in energy range 200-400keV for all quadrants

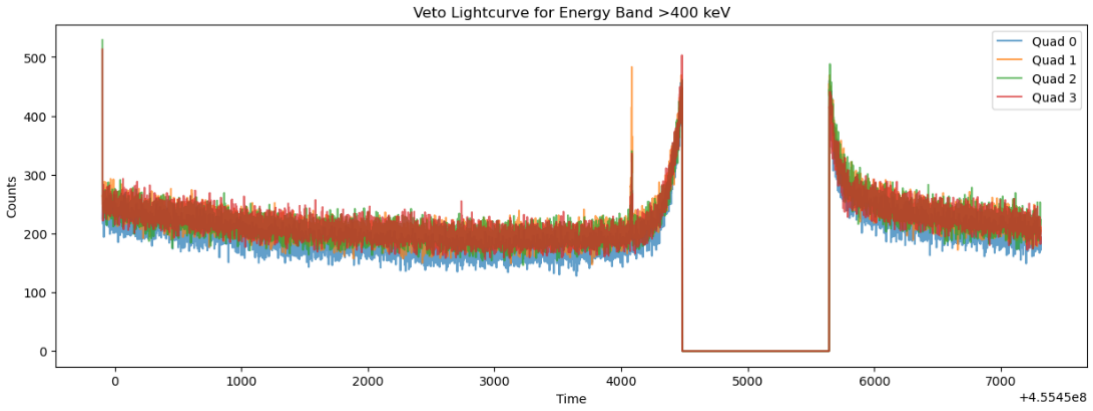


Figure 25: Veto curve in energy range >400keV for all quadrants

We notice that veto doesn't have any counts in the 0-50keV energy band. This is because the Veto detector shows no counts in the 0–50 keV range because it is designed to detect high-energy charged particles and is shielded from low-energy X-rays. As a result, photons below 50 keV cannot reach the Veto detector or fall below its detection threshold. The 100-400 keV energy band in the Veto detector showed large, transient spikes, consistent with SAA-induced background. These features are characteristic of instrumental background and are effectively flagged by the Veto system later on.

2.4 Identification of bangs

Now we move on to define an algorithm that helps us identify if there is a GRB present in our dataset quantitatively and help distinguish it from other transients.

2.4.1 N-sigma algorithm

Now that we have our light curve ready to detect the GRBs and small signals, we go on to identify the potential signals that might be present in our curve. We use an algorithm known as the "N-sigma algorithm" to detect the peaks of the curve. We begin with our detrended data and calculate a threshold for it. We define the threshold as the sum of the mean and n times the standard deviation of our detrended curve. We can change the value of n to set a looser or stricter boundary for the detection of these peaks. We use the in-built function find_peaks from scipy.signal [17]. This function requires the use of two parameters in general; one is the curve that we are finding the peaks of and the other is the prominence. Prominence is the parameter that we use to measure how much a peak stands out from its surroundings and is defined as the vertical distance between the peak and its lowest contour line. It is ideal to use while working with astrophysical transients and other signals. For the detection of peaks for now we use the value of n as 4. We set the prominence as the threshold calculated using this and plot all these peaks along with the detrended data to get an idea about the number of signals we notice in our data. We get the following curve:

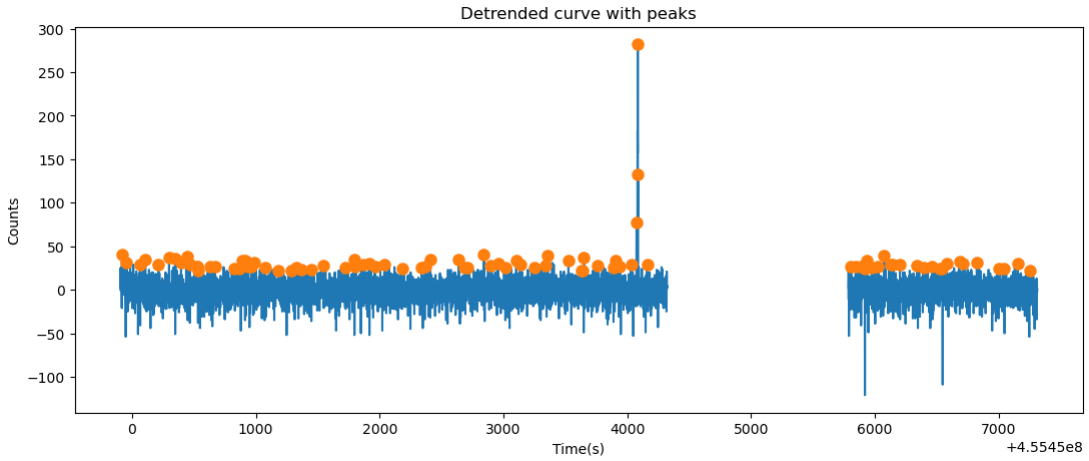


Figure 26: N-sigma algorithm identifying peaks on detrended curve

2.4.2 Detection of potential GRBs

Now we move on to the main part of this project that analyzes the main GRBs detected. We follow a pipeline as given below:

1. Masking and detrending
2. Optimizing time bins

3. Quadrant wise coincidence of peaks (atleast 3)

4. Applying n-sigma to sort out blips from bangs

We already saw the results of the curve from the first step and we figured out that using the time bin of 1s would be ideal. Next we go on to see a minimum 3-quadrant coincidence of peaks. This is a crucial and the most important part of identifying the presence of a GRB. This is because space-based detectors face a lot of non-astrophysical noise: cosmic rays, charged particle events, SAA-induced noise, electronics glitches, etc. Many of these events affect only one quadrant or at most two due to localized interactions. True astrophysical sources like GRBs (Gamma-Ray Bursts), especially bright ones, are simultaneously visible in multiple quadrants and are consistent in timing and spectrum across them.
We set up a condition for the peaks in at least 3 quadrants to be within 1 second time window to be counted as the same peak. We then classify this region around which the peaks coincide to be a potential GRB. First we go on to plot the peaks of all the quadrants.

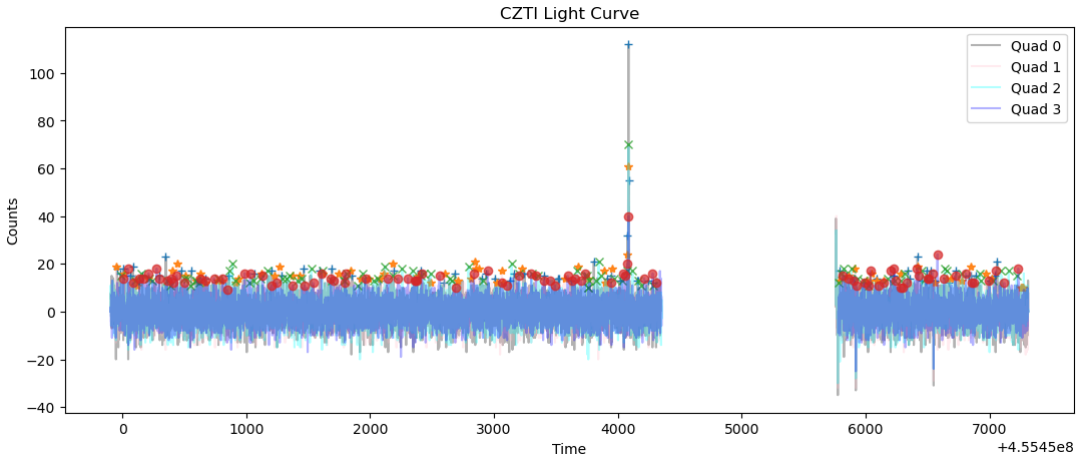


Figure 27: Peaks of the curve from all different quadrants

We see a fair distribution of peaks across the four quadrants. Now we go on to identify if any three peaks belonging to three distinct quadrants lie within the time frame of 10 seconds to each other. If this condition is satisfied by any three peaks then we classify them as coincident and separately mark them out. We now apply this logic to find out the windows where these potential GRBs might exist. We draw a window around their peak to get an idea about their position.

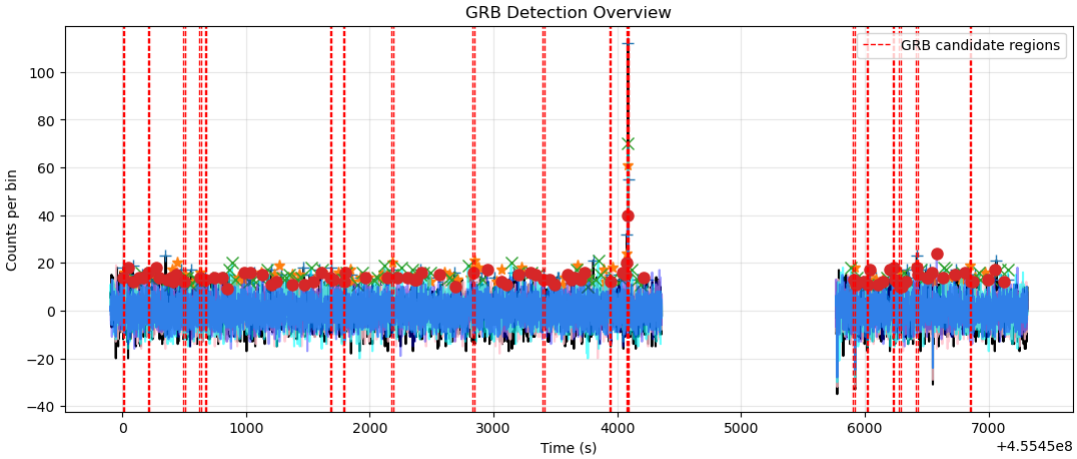


Figure 28: Potential GRB detected marked along with their windows

We find a total of 19 GRB candidates this way. These candidates can either be a small GRB, a long GRB or simply a transient or blip. We now look at the various spectrograms of these detected candidates to see if we can detect a GRB.

2.4.3 Separating GRBs from other transients

Now to move further and detect which one is a GRB in itself, we apply another n-sigma algorithm on these windows and their background. We set up a condition that for the detected signal to be an actual GRB signal, the peak of the window should be higher than the sum of the mean of the background and n times the standard deviation of the background. We identify the background as the total curve minus the detected windows that are present and take the value of n as 4. We get the following results from this.

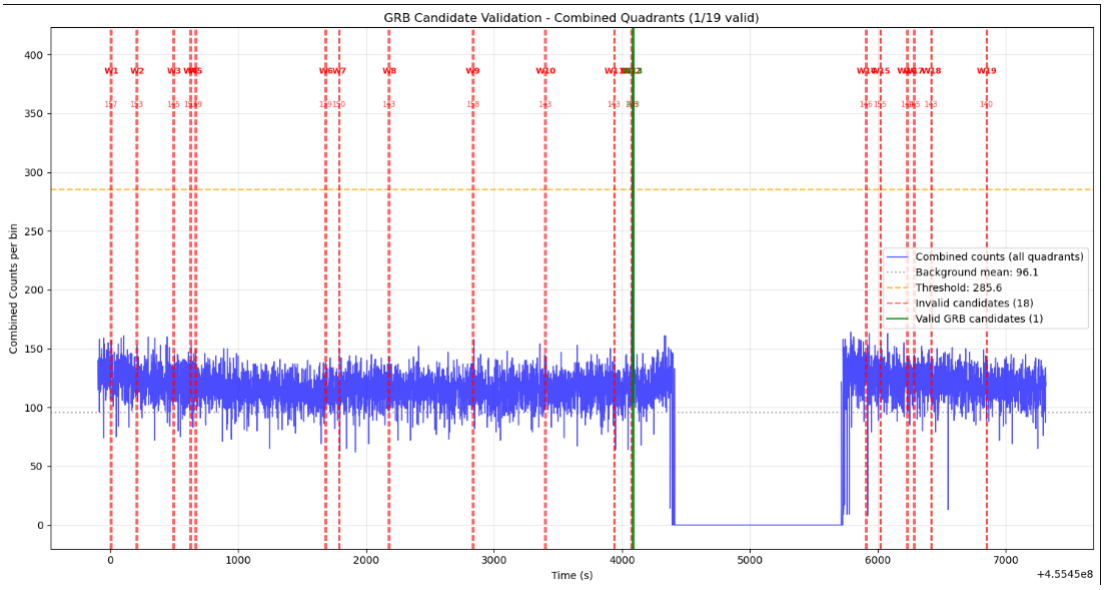


Figure 29: Validated GRB with n-sigma algorithm

We now move onto the spectrogram analysis to analyze the existence of the detected GRB. This would give us the confirmed existence of a GRB.

2.4.4 Spectrogram analysis

We now look at the spectrograms created by these windows. We look at the energy vs time profile while analysing this spectrogram. To do this we first take the windows of the detected events and then define a histogram bin for each time and energy in that window. We do this using the function linspace.
We use the numpy.linspace function to generate evenly spaced numerical values over a specified interval. This is useful for defining time bins or energy ranges with uniform spacing. The function allows precise control over the number of points and whether the endpoint is included [18].
This divides the time and energy range into 150 bins each and we get a 150×150 grid for the spectrogram. We then use the function spectrogram to create the spectrogram using the time and energy bins obtained above and values which are counts of events in each (time, energy) bin. We then draw a 2D histogram as an image.
Since the windows used in this scene are 10s long, we go on to check if one GRB exists in two separate windows or not. We do this by setting plotting spectrograms of all these windows separately.

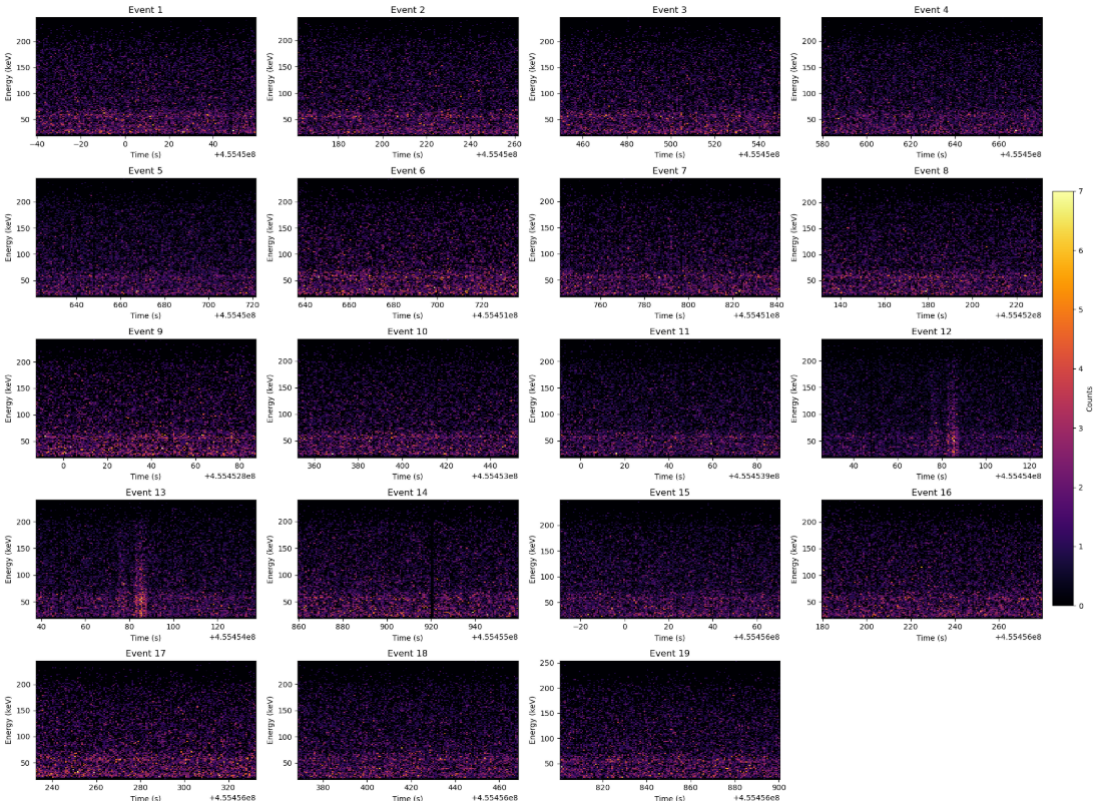
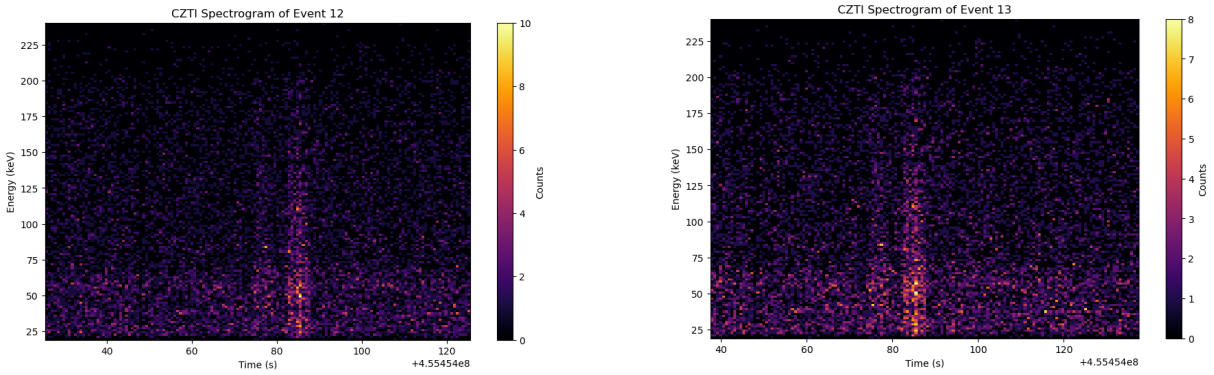


Figure 30: Spectrograms of all the windows

This confirms the presence of our GRB nicely. We now go on to analyze the windows 12 and 13 to take a close look at this. We do this by analysing their spectrograms:



(a) Window 12 spectrogram

(b) Window 13 spectrogram

Figure 31: Spectrograms of two different windows lying side by side

We clearly detect the presence of a GRB around these time windows. This concludes our detection of a GRB. We now look at further analysis of another window of size 100 around this detected peak.

2.4.5 Detailed analysis

Out of all the potential signals detected, we figured out the one which is likely our GRB. We now look at this region more closely and analyse its behavior. Firstly, we plot a quadrant wise graph to see the count distribution from each quadrant. We take a 50 seconds bin around the peak on each side and then plot its counts.
This tells us how the burst behaves across different quadrants. We also see a low background fluctuation which confirms the presence of a GRB. There is no evidence of multiple pulses, so it must be a clean, single-episode GRB. We now look at the energy band analysis to further determine the hard and soft regions. We now plot the same window in different energy bands with different quadrants. We obtain the following results:

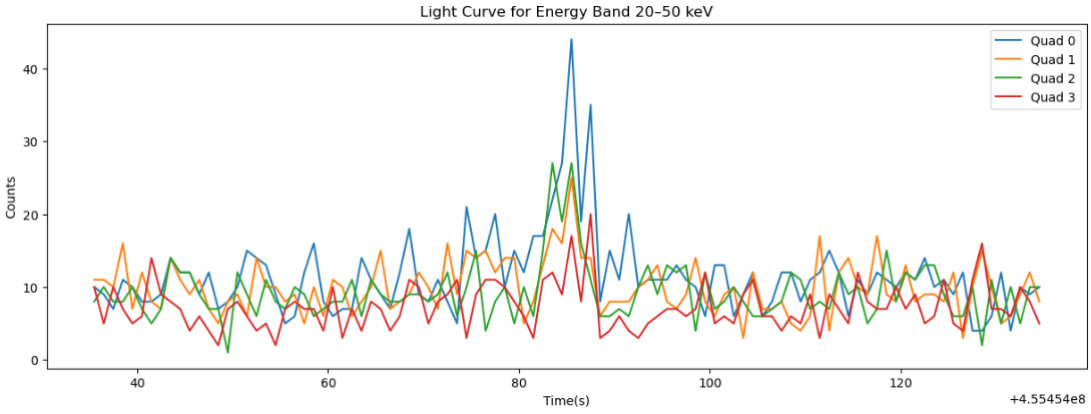


Figure 32: Quadrant wise energy profile for band 20-50keV

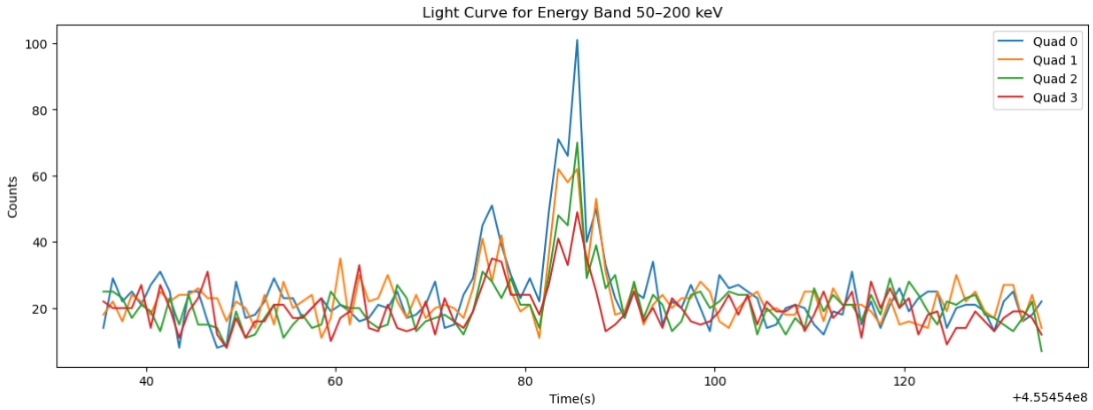


Figure 33: Quadrant wise energy profile for band 50-200keV

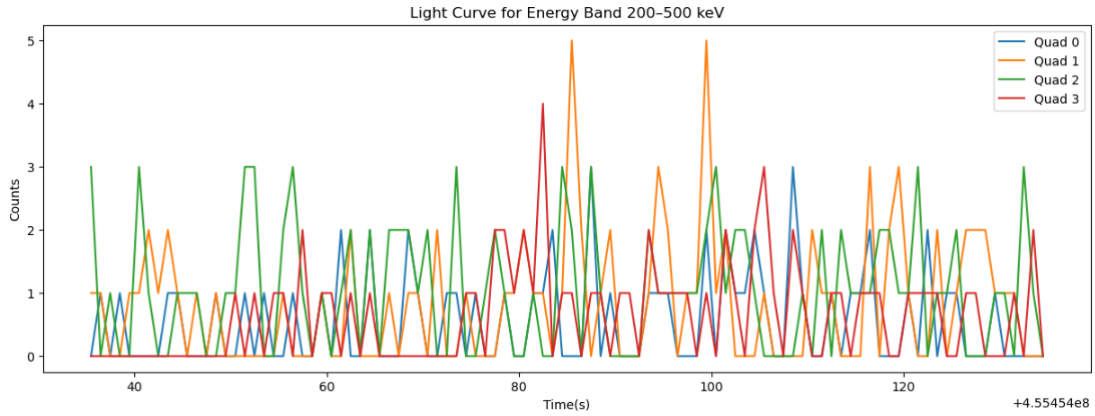


Figure 34: Quadrant wise energy profile for band 200-500keV

We see that the peak in the 50–200 keV band is much more prominent than in the 20–50 keV band. This suggests that the burst is harder in spectrum, which means that more energetic photons were emitted than soft ones. The detection efficiency drops off sharply after 200 keV for many X-ray instruments. So even if photons are arriving, very few are detected.

2.4.6 Veto analysis

To further analyze the origin of the detected burst, it is useful to examine the corresponding Veto detector data in the same energy bands. If the Veto detectors show no significant counts during the burst, it supports the interpretation of a genuine GRB. Conversely, strong simultaneous Veto counts may indicate a background origin or an exceptionally bright GRB capable of penetrating shielding.

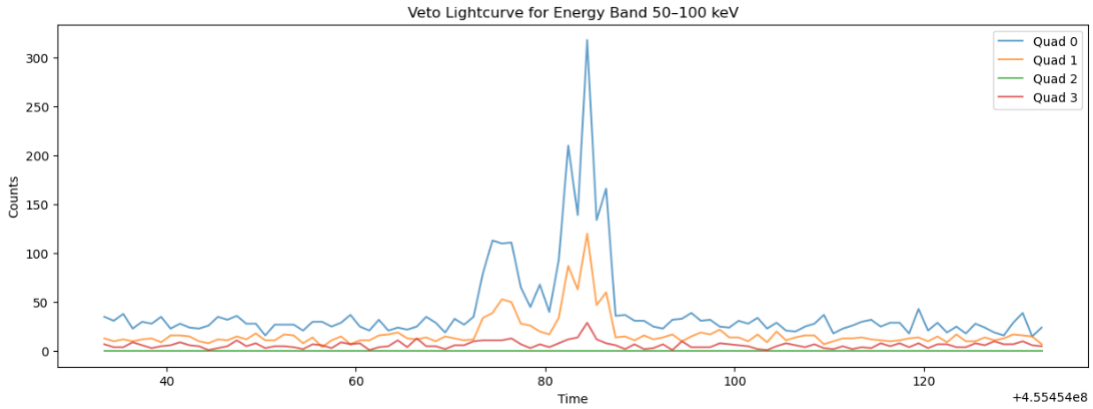


Figure 35: Quadrant wise energy profile for band 50-100keV

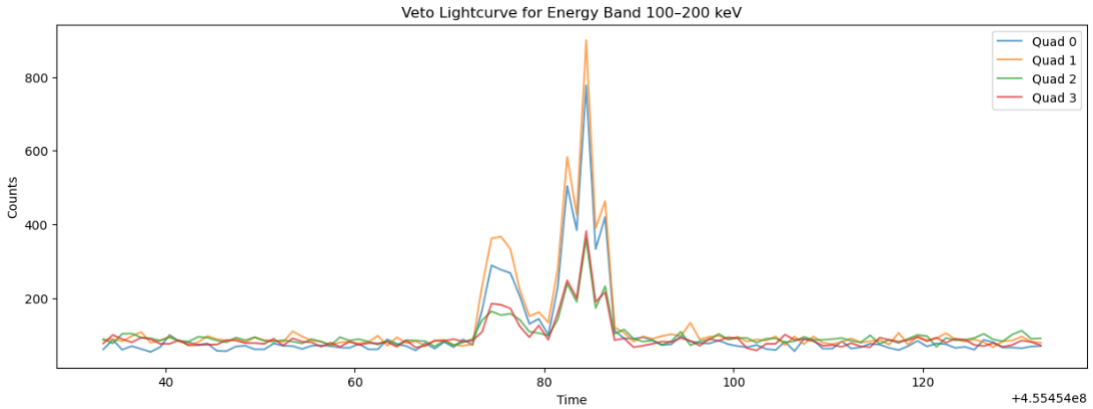


Figure 36: Quadrant wise energy profile for band 100-200keV

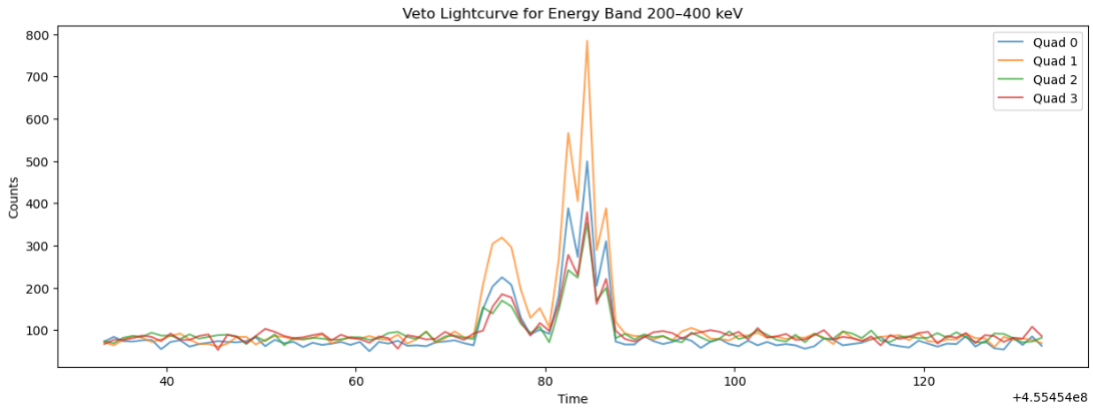


Figure 37: Quadrant wise energy profile for band 200-400keV

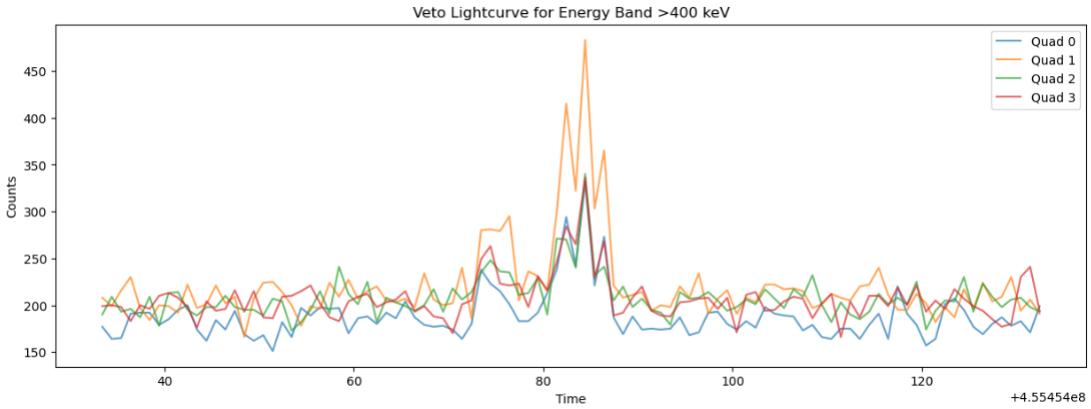


Figure 38: Quadrant wise energy profile for band >400keV

We notice that the Veto detector showed a strong response coinciding with the burst detected by the main CZTI quadrants, particularly in the energy band 100-400 keV. This suggests that the event was exceptionally bright, with high-energy photons capable of penetrating the shielding around the Veto detector. While the Veto is primarily used to flag charged particle backgrounds, such a signal does not necessarily indicate a background event in this case. Instead, the simultaneous detection across both the CZTI and Veto detectors supports the classification of the event as a bright, hard-spectrum GRB.

2.5 Full Algorithm

Now that we have seen all the cases and how we approach our file to find a GRB, let us look at the final pipeline and the key findings of this report.

2.5.1 Final pipeline

The GRB detection and classification algorithm developed in this project follows a structured pipeline designed to identify genuine gamma-ray bursts and distinguish them from background events or instrumental artifacts. The algorithm begins with raw FITS data and processes it through several key stages:

- **Pre-processing** We bin the event data into 1-second intervals and mask it of SAA gaps then smooth it using a median filter.
- **Detrending** We move on with the removal of long-term trends to highlight transient events.
- **Peak Detection** We apply the n-sigma algorithm to detect significant peaks in the detrended lightcurve.
- **Quadrant Coincidence** We then identify the potential GRBs via simultaneous peak detection in at least three CZTI quadrants and then making sure of them by applying a stricter threshold for n-sigma.

- **Spectrogram Analysis** We do a visual inspection of time-energy structure to validate GRB-like signatures.
- **Veto Detector Check** Lastly, we do the analysis of Veto detector data to assess background contamination or confirm bright GRB signatures.

This modular algorithm allows for an efficient detection of GRBs, while minimizing false positives from non-astrophysical sources. We now move on to look at the final results we obtained from this algorithm.

2.5.2 Final results

We mainly first looked at the number of peaks detected when we ran the n-sigma algorithm first using n=4.

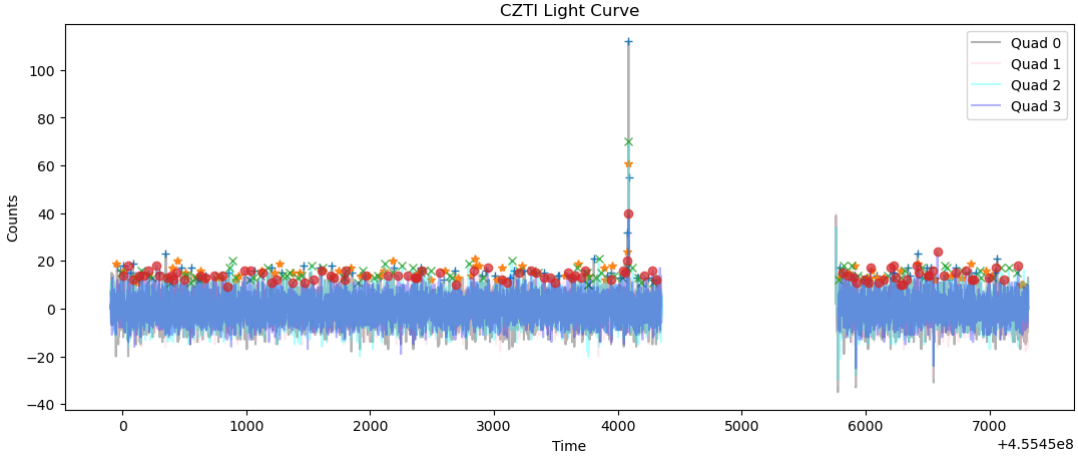


Figure 39: Peaks detected for all quadrants using n=4.

Then we applied the coincidence filter to get the filtered potential GRBs within the graph. We plotted the windows which occured around these detected signals.

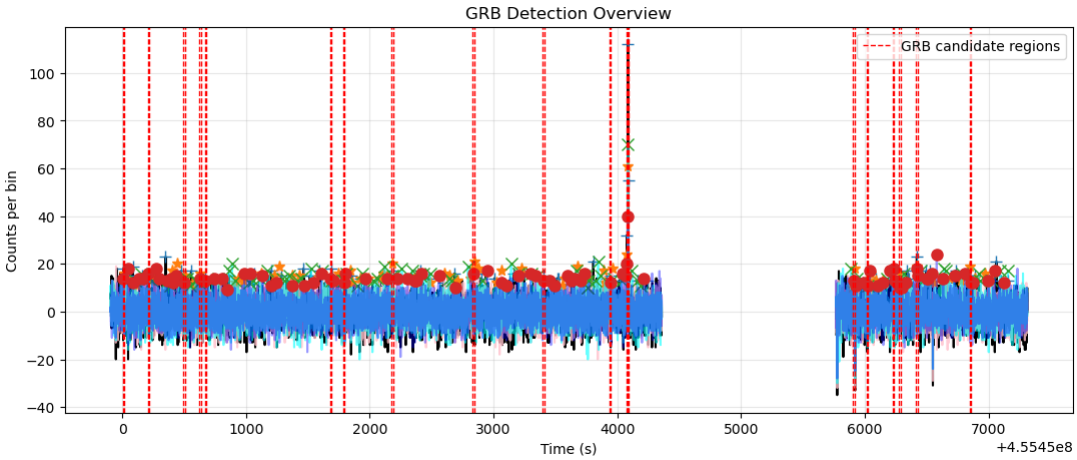


Figure 40: Coincidence filter applied to detect potential GRBs.

Now, we apply n-sigma again with a tighter threshold taking the windows as signals and the curve outside these signals as the background. We take the ones which pass the threshold barrier. We do this to confirm the existence of our GRB.

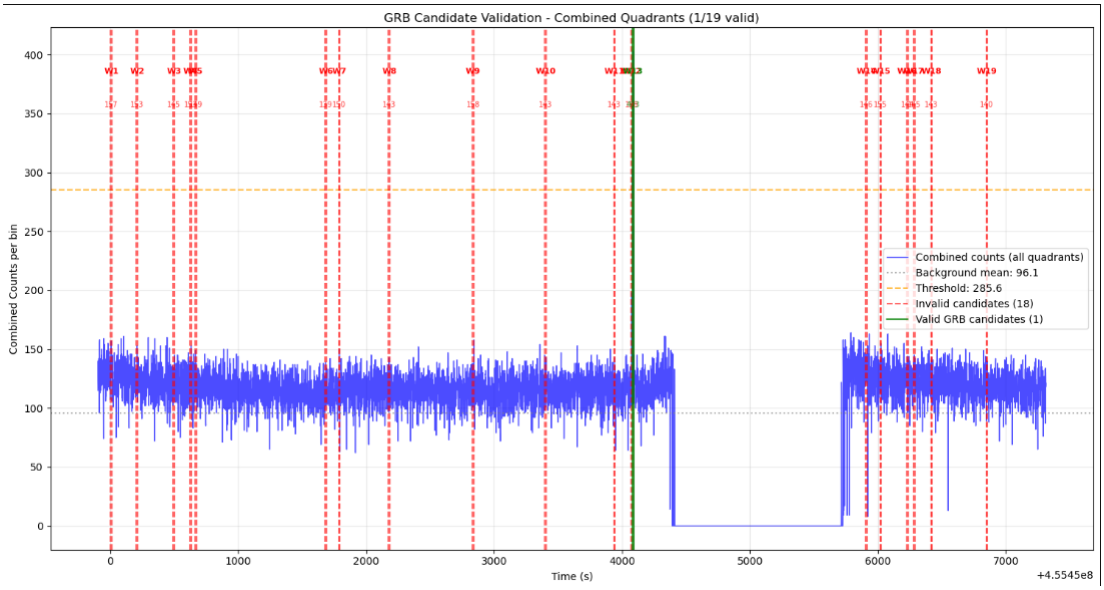


Figure 41: Window analysis for all quadrants with a threshold of n=3.

We get the following results:

```
DETAILED VALIDATION REPORT
================================================================================
Window   Time Range           Peak     Excess   Status   Quadrants
--------------------------------------------------------------------------------
W13      455454082.5-455454092.5s 403.0    6.5      σ VALID    0,1,2,3
W12      455454070.5-455454080.5s 198.0    2.2      σ INVALID  0,1,2,3
W9       455452832.5-455452842.5s 158.0    1.3      σ INVALID  0,1,3
W1       455450004.5-455450014.5s 157.0    1.3      σ INVALID  0,1,3
W15      455456015.5-455456025.5s 155.0    1.2      σ INVALID  0,2,3
W2       455450206.5-455450216.5s 153.0    1.2      σ INVALID  1,2,3
W4       455450624.5-455450634.5s 153.0    1.2      σ INVALID  0,1,2,3
W7       455451786.5-455451796.5s 150.0    1.1      σ INVALID  0,2,3
W5       455450666.5-455450676.5s 149.0    1.1      σ INVALID  0,1,3
W14      455455904.5-455455914.5s 146.0    1.1      σ INVALID  0,1,3
W3       455450494.5-455450504.5s 145.0    1.0      σ INVALID  1,2,3
W17      455456277.5-455456287.5s 145.0    1.0      σ INVALID  1,2,3
W16      455456224.5-455456234.5s 144.0    1.0      σ INVALID  0,2,3
W8       455452176.5-455452186.5s 143.0    1.0      σ INVALID  0,1,2
W10      455453397.5-455453407.5s 143.0    1.0      σ INVALID  0,1,3
W11      455453935.5-455453945.5s 143.0    1.0      σ INVALID  0,1,3
W18      455456413.5-455456423.5s 143.0    1.0      σ INVALID  0,1,3
W19      455456845.5-455456855.5s 140.0    0.9      σ INVALID  0,1,3
W6       455451681.5-455451691.5s 139.0    0.9      σ INVALID  0,2,3

Found 1 valid GRB candidates!
```

Figure 42: Validation report for all the windows detected.

Lastly, we go on to analyze the spectrograms of the windows selected from the above procedure.

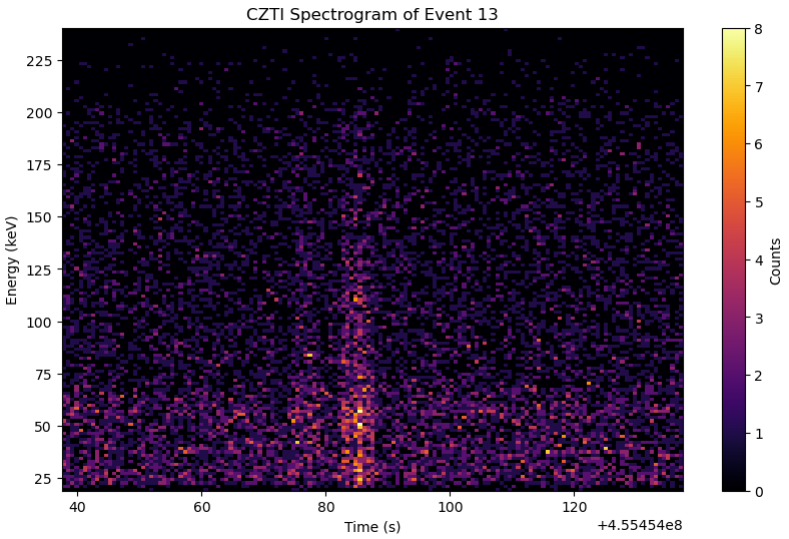


Figure 43: Spectrogram of another window taken around the detected GRB peak

To summarise: Applying the above algorithm to the dataset, a total of 19 GRB candidates were initially detected using the quadrant coincidence method. After applying the n-sigma background validation, one event was confirmed as a genuine GRB.
Spectrogram analysis further validated this event, showing a clear time-energy signature consistent with GRB emission. The energy distribution revealed a hard spectrum, with peak counts in the 50–200 keV band.
Interestingly, the Veto detector also recorded significant counts during the event, particularly in the 100-400 keV band. This suggests that the GRB was very bright, with high-energy photons capable of penetrating the Veto shielding.

3 Conclusion

In this project, we developed and implemented a comprehensive algorithm to detect and validate gamma-ray bursts (GRBs) from CZTI event data. Through a combination of data preprocessing, statistical peak detection, quadrant coincidence analysis, and spectrogram evaluation, we were able to isolate potential GRB candidates and distinguish them from instrumental or background-induced events. We also looked at deeper analysis of the curve by looking at the energy and counts of the curve over time distributed across all quadrants.

The final algorithm effectively filtered out false positives and confirmed the presence of a genuine GRB, characterized by a strong, hard-spectrum signal observed consistently across all CZTI quadrants. Detailed energy analysis revealed that the burst exhibited peak intensity in the 50–200 keV range, consistent with typical GRB profiles.

Overall, this study demonstrates the effectiveness of combining statistical techniques with instrument-specific knowledge to reliably detect GRBs and highlights the critical role of multi-detector validation in high-energy astrophysical event analysis.

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